

ITEC 204 – Data Structures and Algorithms

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- **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: This program is designed to manage IT automation incident tickets using a linear data structure called a list. It allows the user to add, display, search, remove, and count active incident tickets generated by automation bots. The program demonstrates how a list can be used to store, access, and manage data efficiently in a practical IT scenario.

- **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

Code Line: `tickets.append({...})` inside the `add_ticket()` function.

Explanation: The program adds a new incident ticket to the list using the `append()` method. Before adding the ticket, it checks whether the Incident ID already exists by performing a linear search through the list. If the Incident ID is unique, the ticket is successfully stored at the end of the list; otherwise, the program displays a message that the ticket already exists.

Display Incidents:

Code Line: `for ticket in tickets:` inside the `display_tickets()` function.

Explanation: The program displays all active incident tickets by traversing the list using a for loop. Each ticket's Incident ID, Bot Name, and Description are printed in a formatted table for easier reading. If there are no incident tickets stored in the list, the program informs the user that there are no active incident tickets to display.

Search Incident:

Code Line: `for index, ticket in enumerate(tickets):` inside the `search_ticket()` function.

Explanation: The program searches for a ticket based on the Incident ID entered by the user. It performs a linear search by checking each ticket one at a time until it finds a matching Incident ID or reaches the end of the list. If the ticket is found, the complete ticket information is displayed; otherwise, the program notifies the user that no matching ticket exists.

Remove Resolved Incident:

Code Line: `tickets.pop(index)` inside the `remove_ticket()` function.

Explanation: The program first searches for the specified Incident ID using a linear search. Once the ticket is found, it removes the ticket from the list using the `pop()` method, indicating

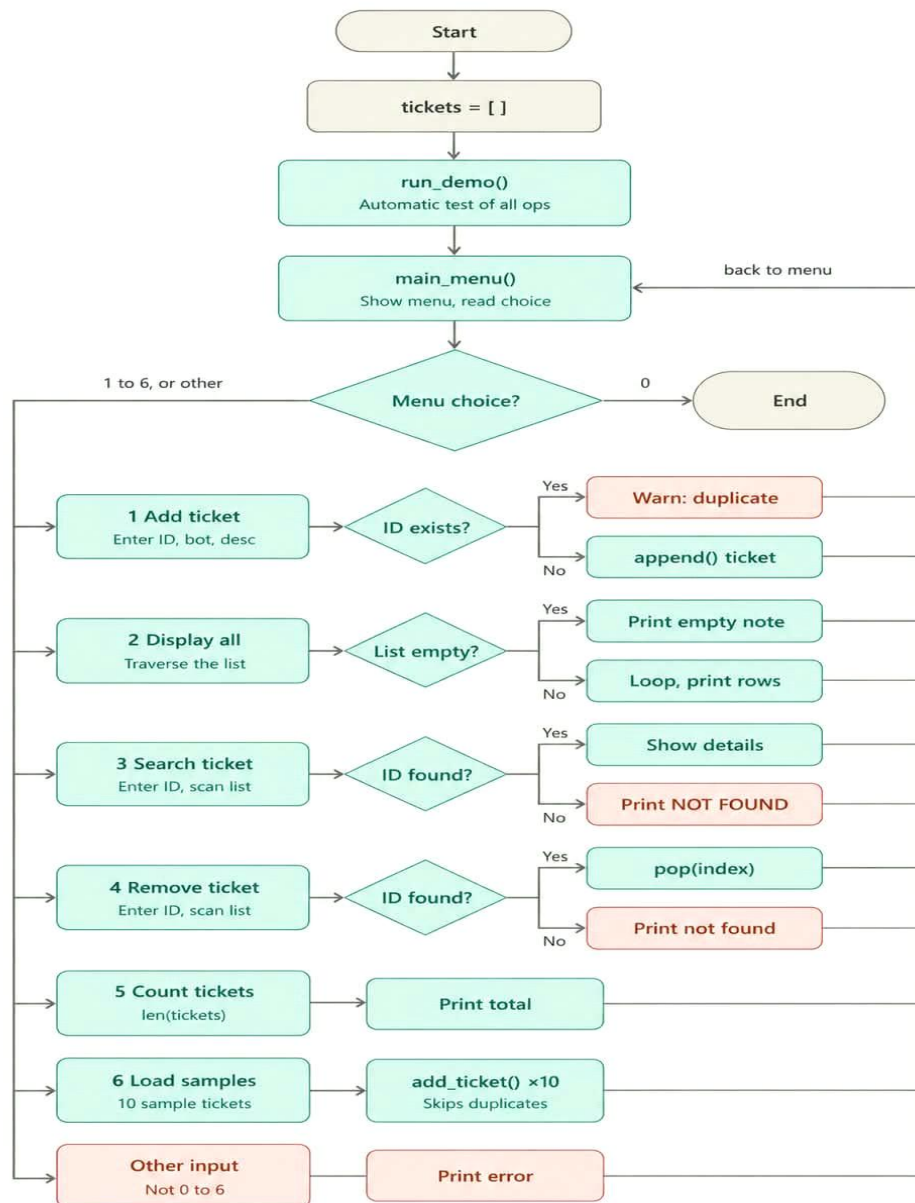
that the incident has already been resolved. If the Incident ID cannot be found, the program displays an error message and no changes are made to the list.

Count Active Incidents:

Code Line: `len(tickets)` inside the `count_active_tickets()` function.

Explanation: The program counts the total number of active incident tickets by using Python's built-in `len()` function. It then displays the current number of unresolved incident tickets stored in the list. This operation helps users quickly monitor how many active incidents still need attention.

Flowchart: Create a flowchart showing the overall process and logic of your program.



- **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

-In this laboratory activity, I learned how a list can be used as a linear data structure to store and organize data efficiently. I also learned how to perform basic operations such as adding, displaying, searching, removing, and counting data using different Python functions and loops. This activity helped me understand the practical application of linear data structures in managing IT automation incident tickets and improved my programming and problem-solving skills.